

Thesis Talk

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Overview of Topics

- Context
- Theoretical Introduction (sorry)
- Overview of Models
- Results of Cohesive Zone Model
- Discussion

What am I doing and why is this relevant?

- Hang out in my office and have coffee (please come by)
- Trying to make an accurate model for crack propagation in snow
- Cracks in snow on large scales can lead to avalanche release
- Average of 24 deaths by avalanche in Switzerland per year, multiple hundred non-fatal accidents (SLF, 2025)
- Understanding relevant because avalanches endanger human life and infrastructure (Schweizer et al., 2003)
- **Ultimate Goal:** determining whether slab breakoff can be seen in DAS data and how this would look

Words I will probably say a lot explained I

- Slab: Part of the snowpack which breaks off and slides downhill (the avalanche itself, I'm looking at dry-snow slab avalanches) (Schweizer et al., 2003)
- Weak layer: Layer with different mechanical properties than the rest of the snow, cracks usually propagate in this layer because it's weaker than the rest of the snowpack (Schweizer et al., 2003)
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Bibliography I



Schweizer, J., Bruce Jamieson, J., & Schneebeli, M. (2003). Snow avalanche formation. *Reviews of Geophysics*, 41(4). <https://doi.org/10.1029/2002RG000123> 



SLF. (2025). Avalanche Victims since 1936 - Long Term Statistics.



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